# Product Description

| Product Information | |
|---|---|
| Product name | Constant Fraction Discriminator |
| Product number | CFD-X02-A |
| Manufacturer | FLIM LABS S.r.l. |
| Registered Address | Via della Farnesina, 3 – 00135 Rome (RM), Italy |




## General Description

The Constant Fraction Discriminator (CFD-X02-A) is a compact, high-performance electronic module designed for precise timing discrimination of fast analog pulses in photon detection, nuclear electronics, spectroscopy, and time-resolved microscopy applications.

It converts analog input signals (e.g., from PMTs, APDs, or pulsed laser synchronization outputs) into well-defined digital timing signals. By applying the constant fraction discrimination principle, the device determines pulse arrival time independently of amplitude, significantly reducing timing walk.

The module features:

- Single-channel architecture
- Four simultaneous digital outputs
- Two SMA outputs (2.5 V logic, 50 Ω)
- Two USB Type-C LVDS outputs
- Selectable polarity switch
- Adjustable output pulse width (5 ns – 250 ns)
- Delay adjustment via external SMA coaxial delay cable
- Compact aluminum enclosure suitable for system integration
- USB Type-C power supply

# Intended Use and Misuse

## Intended Use

The device is intended as a compact electronic module for converting analog to digital timing signals by professional users in research laboratories, universities, or industrial environments. It may be used as a stand-alone unit or integrated into custom measurement systems.

## Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components
- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

# Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°C to +40°C
- Storage temperature: −20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases. Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.